A - i i's

Time Limit: 2 sec / Memory Limit: 1024 MB

Score : 400 points

Problem Statement

You are given a positive integer N. Print an integer sequence of length $L := N(N+1)/2$, $A = (A_1, A_2, \ldots, A_L)$, that satisfies all of the following conditions.

- A contains exactly i occurrences of i for every $i = 1, 2, \ldots, N$.
- $1 \leq |A_i - A_{i+1}| \leq 2$ for every $i = 1, 2, \ldots, L$. Here, A_{L+1} means A_1.

It can be proved that, under the Constraints of this problem, there is always an integer sequence A of length L that satisfies all of the above conditions.

Constraints

- $3 \leq N \leq 1000$
- N is an integer.

Input

The input is given from Standard Input in the following format:

N

Output

Print the elements of A, separated by spaces, in the following format:

$A_1 \quad A_2 \quad \ldots \quad A_L$

If multiple integer sequences A of length L satisfy the conditions in the problem statement, any one of them is accepted.

Sample Input 1

```
4
```

Sample Output 1

```
1 3 4 2 4 3 4 2 4 3
```

The integer sequence $A = (1, 3, 4, 2, 4, 3, 4, 2, 4, 3)$ contains exactly one 1, two 2s, three 3s, and four 4s, satisfying the first condition. The second condition is also satisfied, as follows:

- $|A_1 - A_2| = |1 - 3| = 2$
- $|A_2 - A_3| = |3 - 4| = 1$
- $|A_3 - A_4| = |4 - 2| = 2$
- $|A_4 - A_5| = |2 - 4| = 2$
- $|A_5 - A_6| = |4 - 3| = 1$
- $|A_6 - A_7| = |3 - 4| = 1$
- $|A_7 - A_8| = |4 - 2| = 2$
- $|A_8 - A_9| = |2 - 4| = 2$
- $|A_9 - A_{10}| = |4 - 3| = 1$
- $|A_{10} - A_1| = |3 - 1| = 2$

B - Red and Blue Spanning Tree

Time Limit: 2 sec / Memory Limit: 1024 MB

Score : 600 points

Problem Statement

You are given a connected undirected graph G with N vertices and M edges. For each $i = 1, 2, \ldots, M$, the i-th edge connects vertices a_i and b_i, and is colored red if $c_i =$ R and blue if $c_i =$ B.

Determine if there is a spanning tree of G that satisfies the following condition, and if so, display such a tree.

- For every $i = 1, 2, \ldots, N$,
  - if $s_i =$ R, at least one red edge has vertex i as its endpoint;
  - if $s_i =$ B, at least one blue edge has vertex i as its endpoint.

Constraints

- $2 \leq N \leq 2 \times 10^5$
- $N - 1 \leq M \leq 2 \times 10^5$
- $1 \leq a_i < b_i \leq N$
- c_i is R or B.
- $(a_i, b_i, c_i) \neq (a_j, b_j, c_j)$ if $i \neq j$.
- The given graph is connected.
- s_i is R or B.
- N, M, a_i, b_i are integers.

Input

The input is given from Standard Input in the following format:

$$
\begin{aligned}
&N \quad M \\
&a_1 \quad b_1 \quad c_1 \\
&\vdots \\
&a_M \quad b_M \quad c_M \\
&s_1 s_2 \ldots s_N
\end{aligned}
$$

Output

If no spanning tree of G satisfies the condition, print `No`.

```
No
```

Otherwise, print such a tree in the following format:

```
Yes
t_1 t_2 ... t_{N-1}
```

This means that the i-th edge of the spanning tree is the t_i-th edge of G.
If multiple spanning trees of G satisfy the condition, any one of them is accepted.

Sample Input 1

```
3 3
1 2 R
1 3 B
2 3 B
RRB
```

Sample Output 1

```
Yes
2 1
```

Here we show that the spanning tree formed by the first and second edges of G satisfies the condition.

- $s_1 =$ R, so for $i = 1$, at least one red edge must have vertex 1 as its endpoint. The first edge of G is such an edge.
- $s_2 =$ R, so for $i = 2$, at least one red edge must have vertex 2 as its endpoint. The first edge of G is such an edge.
- $s_3 =$ B, so for $i = 3$, at least one blue edge must have vertex 3 as its endpoint. The second edge of G is such an edge.

Sample Input 2

```
3 4
1 2 R
1 2 B
1 3 B
2 3 B
RRR
```

Sample Output 2

```
No
```

Sample Input 3

```
8 16
5 7 B
2 7 R
1 6 R
1 4 R
6 7 R
4 6 B
4 8 R
2 3 R
3 5 R
6 7 B
2 6 B
5 6 R
1 3 B
4 5 B
2 7 B
1 8 B
BRBRRBRB
```

Sample Output 3

```
Yes
1 2 4 9 11 13 16
```

Sample Input 4

```
8 10
1 7 R
1 3 B
2 5 B
2 8 R
1 5 R
3 6 R
2 6 B
3 4 B
2 8 B
4 6 B
RRRBBBRB
```

Sample Output 4

```
No
```

C - Erase and Divide Game

Time Limit: 2 sec / Memory Limit: 1024 MB

Score : 900 points

Problem Statement

Takahashi and Aoki will play the following game.

1. For each $i = 1, 2, \ldots, N$ in this order, do the following.
    - Write each integer from l_i through r_i once on the blackboard. (Here, l_i and r_i are non-negative integers from the input.)
2. While one or more integers are written on the blackboard, the players take turns doing the following, with Takahashi going first.
    - Choose and perform exactly one of the following two operations.
        - Delete all even numbers written on the blackboard, and replace each remaining integer with half its value rounded down to an integer.
        - Delete all odd numbers written on the blackboard, and replace each remaining integer with half its value.
3. When no integer is written on the blackboard, the game ends and the last player to perform an operation wins.

It can be proved that the game ends in a finite number of operations if Takahashi and Aoki act optimally. Find the winner in this case.

Solve the above problem for T cases.

Constraints

- $1 \leq T \leq 10^4$
- $1 \leq N \leq 10^4$
- $0 \leq l_i \leq r_i \leq 10^{18}$
- $r_i < l_{i+1}$
- The sum of N over the test cases in a single input is at most 10^4.
- All input values are integers.

Input

The input is given from Standard Input in the following format, where test_i represents the i-th test case:

$$
\begin{aligned}
&T \\
&\mathrm{test}_1 \\
&\vdots \\
&\mathrm{test}_T
\end{aligned}
$$

Each test case is given in the following format:

$$
\begin{aligned}
&N \\
&l_1 \quad r_1 \\
&\vdots \\
&l_N \quad r_N
\end{aligned}
$$

Output

Print T lines. The i-th line should contain `Takahashi` if Takahashi wins in the i-th test case, and `Aoki` if Aoki wins.

Sample Input 1

```
3
2
1 2
5 7
1
0 100
10
1312150450968413 28316250877914571
74859962623690078 84324828731963974
148049062628894320 252509054433933439
269587449430302150 335408917861648766
349993004923078531 354979173822804781
522842184971407769 578223540024979436
585335723211047194 615812229161735895
645762258982631926 760713016476190622
779547116602436424 819875141880895723
822981260158260519 919845426262703496
```

Sample Output 1

```
Aoki
Aoki
Takahashi
```

Here is an example of how the game goes for the first test case.

- Each of $1, 2, 5, 6, 7$ is written once on the blackboard.
- Takahashi chooses to delete the odd numbers. $1, 5, 7$ are deleted from the blackboard, and the remaining integers $2, 6$ are replaced by half their values, that is, $1, 3$, respectively.
- Aoki chooses to delete the odd numbers. $1, 3$ are deleted from the blackboard, and no more integer is written on the blackboard, so the game ends and Aoki, who performed the last operation, wins.

D - Red and Blue Chips

Time Limit: 2 sec / Memory Limit: 1024 MB

Score : 900 points

Problem Statement

There are N squares: square 1, square 2, . . ., square N. Each square is colored red or blue. The colors of the squares are represented by a string S of length N. Specifically, for each $i = 1, 2, \ldots, N$, square i is colored red if the i-th character of S is R, and blue if it is B. Square N is always colored blue.

Initially, each square has a chip of the same color as the square.

For each $i = 1, 2, \ldots, N - 1$ in this order, let us perform the following operation.

- Choose an integer j such that $i < j \leq N$ and square j is colored **blue**, and place the pile of chips on square i on top of the pile of chips on square j, keeping the order of the chips.

Print the number of possible sequences of colors of chips of the N-chip pile on square N from top to bottom after the above process, modulo 998244353.

Constraints

- $2 \leq N \leq 300$
- N is an integer.
- S is a string of length N consisting of R and B.
- The N-th character of S is B.

Input

The input is given from Standard Input in the following format:

```
N
S
```

Output

Print the answer.

Sample Input 1

```
4
RBRB
```

Sample Output 1

```
2
```

We use a string consisting of R and B, corresponding to red and blue, to represent the sequence of colors of the chips of a pile from top to bottom. Particularly, the sequence of colors of the chips of a 0-chip pile is an empty string ε.

Also, the state where the sequences of colors of chips of the piles on square $1, 2, 3, 4$ are S_1, S_2, S_3, S_4, respectively, is called as state (S_1, S_2, S_3, S_4).

The following process yields RRBB as the sequence of colors of chips of the pile on square 4.

- First, on each square, place a chip of the same color as the square, resulting in state (R , B , R , B).
- Place the pile of chips on square 1 on top of the pile of chips on square 2, resulting in state (ε, RB , R , B).
- Place the pile of chips on square 2 on top of the pile of chips on square 4, resulting in state (ε, ε, R , RBB).
- Place the pile of chips on square 3 on top of the pile of chips on square 4, resulting in state (ε, ε, ε, RRBB).

The following process yields RBRB as the sequence of colors of chips of the pile on square 4.

- First, on each square, place a chip of the same color as the square, resulting in state (R , B , R , B).
- Place the pile of chips on square 1 on top of the pile of chips on square 4, resulting in state (ε, B , R , RB).
- Place the pile of chips on square 2 on top of the pile of chips on square 4, resulting in state (ε, ε, R , BRB).
- Place the pile of chips on square 3 on top of the pile of chips on square 4, resulting in state (ε, ε, ε, RBRB).

The above two, RRBB and RBRB, are the only possible sequences of colors of chips of the pile on square 4 from top to bottom after the process in the problem.

Sample Input 2

```
20
RRBRRRBBRBBBBRBRBRBB
```

Sample Output 2

```
92378
```

E - Cross Sum Construction

Time Limit: 3 sec / Memory Limit: 1024 MB

Score : 1300 points

Problem Statement

You are given a positive integer N and integer sequences of length N: $A = (a_1, \ldots, a_N)$ and $B = (b_1, \ldots, b_N)$.
Let X be the multiset of N^2 instances $(a_i + b_j)(1 \leq i, j \leq N)$.

For an $N \times N$ matrix M whose elements are integers between -10^{18} and 10^{18}, inclusive, we define the score as follows.

- Let S be the multiset of N^2 instances, each of which is the sum of the $2N - 1$ elements in the i-th row or j-th column of M $(1 \leq i, j \leq N)$. Then, the score is the sum of $\min\bigl($the multiplicity of z in X, the multiplicity of z in $S\bigr)$ over all integers z.

Find a matrix M with the maximum score.

Solve the above problem for T cases.

Constraints

- $1 \leq T \leq 2.5 \times 10^5$
- $1 \leq N \leq 500$
- $-10^9 \leq a_i, b_i \leq 10^9$
- The sum of N^2 over the test cases in a single input is at most 2.5×10^5.
- All input values are integers.

Input

The input is given from Standard Input in the following format, where test_i represents the i-th test case:

$$
\begin{aligned}
&T \\
&\mathrm{test}_1 \\
&\vdots \\
&\mathrm{test}_T
\end{aligned}
$$

Each test case is given in the following format:

$$
\begin{aligned}
&N \\
&a_1 \ \ \ldots \ \ a_N \\
&b_1 \ \ \ldots \ \ b_N
\end{aligned}
$$

Output

For each test case, print a solution in the following format:

$$
\begin{aligned}
&m_{1,1} \ \ \ldots \ \ m_{1,N} \\
&\vdots \\
&m_{N,1} \ \ \ldots \ \ m_{N,N}
\end{aligned}
$$

Here, $m_{i,j}$ is the element at the i-th row and j-th column of a matrix M with the maximum score, and must be an integer such that $-10^{18} \leq m_{i,j} \leq 10^{18}$.
If multiple matrices M have the maximum score, any one of them is accepted.

Sample Input 1

```
3
1
5
-10
2
0 -1
8 -11
3
20 23 26
1 2 3
```

Sample Output 1

```
-5
8 9
-10 -9
2 9 4
7 5 3
6 1 8
```

For the input and output in the first case, $X = \{-5\}, S = \{-5\}$, for a score of 1.
For the input and output in the second case, $X = \{8, -11, 7, -12\}, S = \{7, 8, -11, -10\}$, for a score of 3.
For the input and output in the third case, $X = \{21, 22, 23, 24, 25, 26, 27, 28, 29\}, S = \{28, 21, 26, 23, 25, 27, 24, 29, 22\}$, for a score of 9.

F - No Permutations

Time Limit: 8 sec / Memory Limit: 1024 MB

Score : 1700 points

Problem Statement

You are given a positive integer N. Print the number, modulo 998244353, of sequences A of length $3N$ that contain each integer from 1 through N three times and satisfy the following condition.

- A has no contiguous subsequence of length N that is a permutation of the sequence $(1, 2, \ldots, N)$.

Constraints

- $1 \leq N \leq 2 \times 10^5$
- All input values are integers.

Input

The input is given from Standard Input in the following format:

```
N
```

Output

Print the answer.

Sample Input 1

```
3
```

Sample Output 1

```
132
```

For instance, $A = (1, 3, 3, 2, 2, 2, 1, 1, 3)$ satisfies the condition in the problem statement. On the other hand, $A = (1, 3, 3, 2, 2, 3, 1, 1, 2)$ does not, because the 5-th, 6-th, 7-th elements of A form a contiguous subsequence that is a permutation of the sequence $(1, 2, 3)$.

Sample Input 2

```
123456
```

Sample Output 2

```
31984851
```